Lab-1 & Lab-2

Command	Syntax	Purpose / Description	Example
uname -r	uname -r	Displays the current Linux kernel version.	uname -r
uname -m	uname -m	Shows the system architecture (32-bit/64-bit).	uname -m
pwd	pwd	Displays the current working directory.	pwd
ls	ls	Lists files and directories in the current directory.	ls
ls -a	ls -a	Displays all files including hidden files.	ls -a
mkdir	mkdir dirname	Creates a new directory.	mkdir OS1
mkdir (multiple)	mkdir dir1 dir2 dir3	Creates multiple directories at once.	mkdir foo bar baz
cd	cd path	Changes the current directory.	cd /Desktop/OS
cd	cd	Returns to the home directory.	cd
cd ..	cd ..	Moves to the parent directory.	cd ..
touch	touch filename	Creates an empty file.	touch test.txt
cat >	cat > filename	Creates a new file and allows the user to enter text.	cat > ex1
cat	cat filename	Displays the contents of a file.	cat ex1
cat >>	cat >> filename	Appends new content to an existing file.	cat >> ex1
cat file1 >> file2	cat file1 >> file2	Appends the contents of one file to another.	cat file1.txt >> file2.txt

cat file1 file2 > file3	cat file1 file2 > file3	Concatenates multiple files into a new file.	cat file1 file2 > file3
cp	cp source destination	Copies files or directories.	cp ex1 ex2
mv	mv oldfile newfile	Renames or moves files/directories.	mv ex1 ex3
rm	rm filename	Deletes a file.	rm ex1
rm *	rm *	Deletes all files in the current directory.	rm *
rm -r	rm -r directory	Deletes a directory and all its contents recursively.	rm -r /tmp/backup
rmdir	rmdir dirname	Removes an empty directory.	rmdir flower
rmdir -p	rmdir -p dir1/dir2/dir3	Removes a directory and its empty parent directories.	rmdir -p dir1/dir2/dir3

Additional Must-Know Linux Commands

Command	Syntax	Purpose	Example
clear	clear	Clears the terminal screen.	clear
history	history	Displays previously executed commands.	history
man	man command	Shows the manual/help page for a command.	man ls
echo	echo text	Prints text on the screen or into a file.	echo "Hello"
date	date	Displays the current date and time.	date
whoami	whoami	Displays the current logged-in username.	whoami
who	who	Shows currently logged-in users.	who
hostname	hostname	Displays the system hostname.	hostname
cal	cal	Displays a calendar.	cal

file	file filename	Identifies the type of a file.	file notes.txt
head	head filename	Displays the first 10 lines of a file.	head file.txt
tail	tail filename	Displays the last 10 lines of a file.	tail file.txt
wc	wc filename	Counts lines, words, and characters in a file.	wc notes.txt
sort	sort filename	Sorts lines of a file alphabetically.	sort marks.txt
grep	grep pattern filename	Searches for text inside a file.	grep "Unix" notes.txt
find	find path -name filename	Searches for files/directories.	find . -name test.txt
chmod	chmod permission file	Changes file permissions.	chmod 755 script.sh
sudo	sudo command	Executes a command with administrator privileges.	sudo apt update
apt	apt command	Package management command in Ubuntu.	sudo apt install vim
nano	nano filename	Opens a simple text editor in terminal.	nano test.txt

Commands for Navigation

Command	**Purpose**	**Example**
cd ~	Go to home directory	cd ~
cd -	Return to previous directory	cd -
ls -l	Long listing format	ls -l
ls -lh	Human-readable file sizes	ls -lh
ls -R	Recursively list directories	ls -R